

Solve a Graph using Depth First Search (DFS)

Aim :

To Implement DFS traversal on a graph to explore all reachable nodes from a source.

Procedure:

1. Define the graph using an adjacency list.
2. Start from the source node.
3. Recursively visit all unvisited neighbors.

Python Code:

```
def dfs(graph,start,visited=None):
    if visited is None:
        visited=set()
    visited.add(start)
    print(start,end=" ")
    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph,neighbor,visited)
graph={
    'A':['B','C'],
    'B':['D','E'],
    'C':['F'],
    'D':[],
    'E':['F'],
    'F':[]
}
print("DFS traversal from node A:")
dfs(graph,'A')
```

Output:

```
DFS traversal from node A:
```

```
A B D E F C
```

Result:

Thus the program of implementing DFS traversal on a graph to explore all reachable nodes from a source. was executed successfully.